INHA UNIVERSITY IN TASHKENT (FALL 2024)

Computer Algorithms (CIE3090)

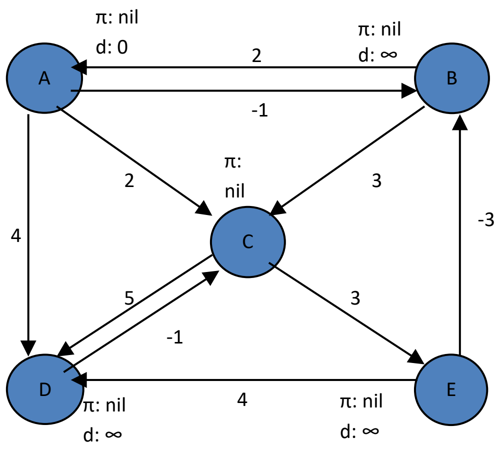
ASSIGNMENT #5

Section No._________ ID___________________________ Name____________________________

Submission Deadline: **28 /Nov/ 2024 (23:30 Hr)**

- Solve all questions on printout of this page. You are required to solve **(handwritten)** all the questions and submit it on eclass by taking picture of your assignment. **NO hardcopy submission is required**

Q1. Find the shortest path from vertex A to all other nodes using Bellman Ford Algorithm



Edges	W(u,v)
A→ B	
A→ D	
A → C	

vertex	Initial d[v]	Itr1 - d[v]	Itr2 - d[v]	Itr3 - d[v]	Itr4 - d[v]	Itr5 - d[v]
A						
B						
C						
D						
E						

Q2. Solve the following Knapsack Problem using Dynamic programming

Weight <2, 3, 4, 5>
Value <13, 7, 20, 9>
Capacity = 5 kg
(use the below table for all calculations, kindly add if more column/row required, similarly leave any extra column or row if exist)

Answer :

Items identified = ________________

Maximum Gain = _________________